

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Power Island Off-design Heat Balances (Case Book)

Document number	IEC-ALP-PER-003
Revision	0
Date	2026-09-28
Status	Issued for EPC engineering (reference data)
Addressee	Istanbul EPC Muhendislik ve Taahhut A.S. (consortium leader) - EPC engineering
Contract	ALP-EPC-001 (Owner contract); consortium agreement IEPC-IEC

	Prepared	Checked	Approved
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Signature / date	signed 2026-09-28	signed 2026-09-28	signed 2026-09-28

Rev	Date	Description
0	2026-09-27	Issued for EPC engineering (reference data)

CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. Purpose and method

Rated off-design heat balances of the IEC power island for the EPC plant heat and mass balance (all operating and guarantee cases agreed in TQ-IEC-017). Expected values, new and clean.

- Design point and surfaces: IEC-ALP-PER-001 Rev B (corrected HRSG arrangement, EPC piping values). The model reproduces the Rev B design point exactly.
- HRSG: fixed surface per section (UA from the design point, gas-side scaling with $\text{flow}^{0.6}$), HP superheater and reheater, HP economiser 1 and IP economiser as parallel banks with the design gas split; economisers limited to 2 K below saturation; HRSG casing loss 0.4 % of gas-side duty.
- Steam turbine: sliding pressure (Stodola cone law) above 60 bar(a) HP; section efficiencies reduced off design ($1 - 0.3 (1 - m/m_{\text{design}})^2$); LP exhaust loss proportional to the square of the exhaust volume flow (minimum 30 %).
- Attemperation to 600 degC main and reheat steam with spray water from the BFP discharge / interstage (streams 29 and 30 where in service).
- Condenser: fixed surface (HEI cleanliness 0.85), constant CW flow as at SRC; seawater cp 4.07 kJ/kg K.
- GT: IE-9H.02 correction curves; output limit 470 MW (generator/shaft) below about -1 degC; part load by IGV (exhaust temperature held at 640 degC down to 60 % load); minimum emissions-compliant load 40 % GT load.
- LDO: C12H23 pseudo-formula, LHV 42.9 MJ/kg, 30 degC at IF-02; water injection 1.02 kg water per kg fuel at 60 bar(a), 30 degC (IF-03); performance gas heater out of service.
- Case SRC-NG-060: GT load found by IEC so that the plant net output is 60 % of case SRC-NG-100 using the EPC auxiliary-load model (DEC-EPCE-0001); the EPC HMB confirms the achieved load.

2. Case summary

Case	Ambient degC	RH %	Seawater degC	Fuel	GT load %	GT MW	GT HR kJ/kWh	Exhaust kg/s	Exhaust degC	ST MW	Gross MW	Stack degC	Condenser mbar	Spray HP/RH kg/s	Water inj. kg/s
SRC-NG-100	15.0	70	16.0	GAS	100.0	425.0	8,314	796.7	640.0	200.73	625.73	80.2	34.4	0.00 / 0.00	-
SRC-NG-GT80	15.0	70	16.0	GAS	80.0	340.0	8,746	696.5	640.0	175.05	515.05	76.1	32.0	1.39 / 0.31	-
SRC-NG-060	15.0	70	16.0	GAS	55.8	237.1	9,629	573.3	636.6	140.73	377.80	71.0	29.2	1.59 / 0.40	-
SRC-NG-MEL	15.0	70	16.0	GAS	40.0	170.0	10,600	496.9	615.0	113.23	283.23	69.1	27.3	0.50 / 0.00	-
SRC-LDO-100	15.0	70	16.0	LDO	100.0	395.0	8,650	715.8	622.0	175.45	570.45	82.8	32.5	0.15 / 0.00	22.57
WIN-NG-100	-4.8	80	8.0	GAS	100.0	470.0	8,131	832.5	635.7	207.94	677.95	77.0	21.7	0.00 / 0.00	-
MIN-NG-100	-8.0	80	6.9	GAS	100.0	470.0	8,089	823.5	634.0	204.92	674.92	76.1	20.1	0.00 / 0.00	-
SUM-NG-100	32.8	45	24.0	GAS	100.0	372.8	8,501	729.5	649.3	184.53	557.33	81.7	52.7	1.52 / 0.68	-
MAX-NG-100	40.0	35	27.0	GAS	100.0	348.2	8,595	695.9	653.0	175.84	524.07	81.9	61.1	2.06 / 0.95	-

GT output at the limit in cases WIN-NG-100 and MIN-NG-100. Gross output before auxiliaries and transformer losses; net values are the EPC's. IEC notes: SRC-NG-MEL (GT 40 %) gives about 45 % plant net load; SRC-LDO-100 about 559 MW net with the EPC auxiliaries - see the consortium clarifications.

3.1 Case SRC-NG-100 - Site Reference Conditions, natural gas, base load (guarantee case)

Ambient 15.0 degC, 70.0 % RH, seawater 16.0 degC; GT 425.0 MW (100.0 % load), HR 8,314 kJ/kWh; ST 200.73 MW; GSU losses GT 1107 kW, ST 584 kW; HRSG gas temperatures 640.0, 450.6, 371.6, 366.9, 295.1, 261.0, 198.7, 258.2, 166.2, 80.2 degC.

No	Description	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	AMB	GT	776.006	1.012	15.00	-
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	FGS	FGH	20.672	38.000	25.00	-
3	Heated fuel gas to GT combustion system	FGH	GT	20.672	35.500	215.00	-
4	GT exhaust gas to HRSG inlet	GT	HRSG	796.678	1.048	640.00	-
5	Flue gas at HRSG outlet to stack	HRSG	STK	796.678	1.014	80.16	-
6	HP steam at HRSG outlet (HP SH outlet)	HRSG	PIPE	100.155	187.000	600.00	-
7	HP steam at ST main stop valve inlet	PIPE	ST	100.155	181.500	598.50	-
8	Cold reheat steam at HP turbine exhaust	ST	PIPE	100.155	40.200	366.21	-
9	Cold reheat steam at HRSG reheater inlet	PIPE	HRSG	100.155	39.500	365.21	-
10	IP superheated steam (joins cold reheat inside the HRSG)	HRSG	HRSG	17.338	40.300	330.00	-
11	Hot reheat steam at HRSG outlet	HRSG	PIPE	117.493	37.600	600.00	-
12	Hot reheat steam at ST IP stop/control valve inlet	PIPE	ST	117.493	36.200	598.50	-
13	LP steam at HRSG outlet	HRSG	PIPE	11.901	5.400	250.00	-
14	LP steam at ST LP admission	PIPE	ST	11.901	5.050	249.00	-
15	IP turbine exhaust (crossover, internal)	ST	ST	117.493	4.950	309.20	-
16	LP turbine exhaust to condenser (UEEP)	ST	COND	129.394	0.03444	26.40	2,355.50
17	Condensate at hotwell outlet / CEP suction	COND	CEP	129.394	0.03444	26.35	-
18	Condensate at CEP discharge	CEP	CMIX	129.394	16.000	26.52	-
19	Fuel gas heater water return to condensate	FGH	CMIX	12.481	43.000	75.00	-
20	Condensate to HRSG condensate preheater inlet	CMIX	HRSG	141.874	15.500	30.85	-
21	Feedwater from LP drum to boiler feed pumps	HRSG	BFP	129.973	5.900	158.13	-
22	HP feedwater at BFP discharge to HP economiser	BFP	HRSG	100.155	209.000	161.83	-
23	IP feedwater at BFP interstage to IP economiser	BFP	HRSG	29.819	46.000	158.89	-
24	IP economiser water to performance fuel gas heater	HRSG	FGH	12.481	43.300	247.98	-
25	Circulating seawater to condenser inlet	SEA	COND	10,196.804	2.500	16.00	-
26	Circulating seawater from condenser outlet	COND	OUT	10,196.804	1.700	23.00	-

3.2 Case SRC-NG-GT80 - Site Reference Conditions, natural gas, GT at 80 % load

Ambient 15.0 degC, 70.0 % RH, seawater 16.0 degC; GT 340.0 MW (80.0 % load), HR 8,746 kJ/kWh; ST 175.05 MW; GSU losses GT 777 kW, ST 471 kW; HRSG gas temperatures 640.0, 457.0, 360.6, 355.9, 288.5, 252.8, 191.9, 250.1, 159.9, 76.1 degC.

No	Description	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	AMB	GT	679.101	1.012	15.00	-
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	FGS	FGH	17.398	38.000	25.00	-
3	Heated fuel gas to GT combustion system	FGH	GT	17.398	35.500	215.00	-
4	GT exhaust gas to HRSG inlet	GT	HRSG	696.499	1.039	640.00	-
5	Flue gas at HRSG outlet to stack	HRSG	STK	696.499	1.014	76.14	-
6	HP steam at HRSG outlet (HP SH outlet)	HRSG	PIPE	87.346	163.084	600.00	-
7	HP steam at ST main stop valve inlet	PIPE	ST	87.346	158.287	598.50	-
8	Cold reheat steam at HP turbine exhaust	ST	PIPE	87.346	35.276	369.63	-
9	Cold reheat steam at HRSG reheater inlet	PIPE	HRSG	87.346	34.670	368.63	-
10	IP superheated steam (joins cold reheat inside the HRSG)	HRSG	HRSG	15.475	35.396	323.12	-
11	Hot reheat steam at HRSG outlet	HRSG	PIPE	103.126	33.002	600.00	-
12	Hot reheat steam at ST IP stop/control valve inlet	PIPE	ST	103.126	31.773	598.50	-
13	LP steam at HRSG outlet	HRSG	PIPE	10.253	4.709	243.95	-
14	LP steam at ST LP admission	PIPE	ST	10.253	4.412	242.95	-
15	IP turbine exhaust (crossover, internal)	ST	ST	103.126	4.337	310.68	-
16	LP turbine exhaust to condenser (UEEP)	ST	COND	113.378	0.03204	25.18	2,366.47
17	Condensate at hotwell outlet / CEP suction	COND	CEP	113.378	0.03204	25.13	-
18	Condensate at CEP discharge	CEP	CMIX	113.378	16.000	25.29	-
19	Fuel gas heater water return to condensate	FGH	CMIX	11.002	37.068	75.00	-
20	Condensate to HRSG condensate preheater inlet	CMIX	HRSG	124.381	15.500	29.74	-
21	Feedwater from LP drum to boiler feed pumps	HRSG	BFP	114.128	5.134	152.79	-
22	HP feedwater at BFP discharge to HP economiser	BFP	HRSG	85.959	180.153	155.90	-
23	IP feedwater at BFP interstage to IP economiser	BFP	HRSG	26.477	40.068	153.44	-
24	IP economiser water to performance fuel gas heater	HRSG	FGH	11.002	38.256	240.73	-
25	Circulating seawater to condenser inlet	SEA	COND	10,196.804	2.500	16.00	-
26	Circulating seawater from condenser outlet	COND	OUT	10,196.804	1.700	22.18	-
29	HP attemperation spray water (BFP discharge)	BFP	HRSG	1.386	180.153	155.90	-
30	Reheat attemperation spray water (BFP interstage)	BFP	HRSG	0.305	40.068	153.44	-

3.3 Case SRC-NG-060 - Site Reference Conditions, natural gas, 60 % plant net load (guarantee PG-03)

Ambient 15.0 degC, 70.0 % RH, seawater 16.0 degC; GT 237.1 MW (55.8 % load), HR 9,629 kJ/kWh; ST 140.73 MW; GSU losses GT 475 kW, ST 343 kW; HRSG gas temperatures 636.6, 461.6, 343.7, 339.0, 280.4, 241.4, 182.2, 238.8, 151.0, 71.0 degC.

No	Description	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	AMB	GT	559.914	1.012	15.00	-
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	FGS	FGH	13.356	38.000	25.00	-
3	Heated fuel gas to GT combustion system	FGH	GT	13.356	35.500	215.00	-
4	GT exhaust gas to HRSG inlet	GT	HRSG	573.270	1.030	636.63	-
5	Flue gas at HRSG outlet to stack	HRSG	STK	573.270	1.014	71.02	-
6	HP steam at HRSG outlet (HP SH outlet)	HRSG	PIPE	71.022	132.605	600.00	-
7	HP steam at ST main stop valve inlet	PIPE	ST	71.022	128.705	598.50	-
8	Cold reheat steam at HP turbine exhaust	ST	PIPE	71.022	29.015	377.29	-
9	Cold reheat steam at HRSG reheater inlet	PIPE	HRSG	71.022	28.528	376.29	-
10	IP superheated steam (joins cold reheat inside the HRSG)	HRSG	HRSG	13.430	29.192	310.90	-
11	Hot reheat steam at HRSG outlet	HRSG	PIPE	84.855	27.155	600.00	-
12	Hot reheat steam at ST IP stop/control valve inlet	PIPE	ST	84.855	26.144	598.50	-
13	LP steam at HRSG outlet	HRSG	PIPE	8.244	3.845	235.08	-
14	LP steam at ST LP admission	PIPE	ST	8.244	3.610	234.08	-
15	IP turbine exhaust (crossover, internal)	ST	ST	84.855	3.562	316.36	-
16	LP turbine exhaust to condenser (UEEP)	ST	COND	93.100	0.02925	23.66	2,396.87
17	Condensate at hotwell outlet / CEP suction	COND	CEP	93.100	0.02925	23.61	-
18	Condensate at CEP discharge	CEP	CMIX	93.100	16.000	23.77	-
19	Fuel gas heater water return to condensate	FGH	CMIX	9.062	29.824	75.00	-
20	Condensate to HRSG condensate preheater inlet	CMIX	HRSG	102.161	15.500	28.36	-
21	Feedwater from LP drum to boiler feed pumps	HRSG	BFP	93.917	4.182	145.17	-
22	HP feedwater at BFP discharge to HP economiser	BFP	HRSG	69.435	144.757	147.58	-
23	IP feedwater at BFP interstage to IP economiser	BFP	HRSG	22.491	32.824	145.69	-
24	IP economiser water to performance fuel gas heater	HRSG	FGH	9.062	31.935	230.21	-
25	Circulating seawater to condenser inlet	SEA	COND	10,196.804	2.500	16.00	-
26	Circulating seawater from condenser outlet	COND	OUT	10,196.804	1.700	21.16	-
29	HP attemperation spray water (BFP discharge)	BFP	HRSG	1.587	144.757	147.58	-
30	Reheat attemperation spray water (BFP interstage)	BFP	HRSG	0.404	32.824	145.69	-

3.4 Case SRC-NG-MEL - Site Reference Conditions, natural gas, GT at minimum emissions-compliant load (40 %)

Ambient 15.0 degC, 70.0 % RH, seawater 16.0 degC; GT 170.0 MW (40.0 % load), HR 10,600 kJ/kWh; ST 113.23 MW; GSU losses GT 337 kW, ST 261 kW; HRSG gas temperatures 615.0, 458.2, 328.1, 323.5, 272.2, 231.1, 175.1, 228.6, 144.2, 69.1 degC.

No	Description	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	AMB	GT	486.356	1.012	15.00	-
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	FGS	FGH	10.543	38.000	25.00	-
3	Heated fuel gas to GT combustion system	FGH	GT	10.543	35.500	215.00	-
4	GT exhaust gas to HRSG inlet	GT	HRSG	496.899	1.026	615.00	-
5	Flue gas at HRSG outlet to stack	HRSG	STK	496.899	1.014	69.10	-
6	HP steam at HRSG outlet (HP SH outlet)	HRSG	PIPE	58.533	109.287	600.00	-
7	HP steam at ST main stop valve inlet	PIPE	ST	58.533	106.072	598.50	-
8	Cold reheat steam at HP turbine exhaust	ST	PIPE	58.533	24.023	384.98	-
9	Cold reheat steam at HRSG reheater inlet	PIPE	HRSG	58.533	23.623	383.98	-
10	IP superheated steam (joins cold reheat inside the HRSG)	HRSG	HRSG	11.950	24.258	299.56	-
11	Hot reheat steam at HRSG outlet	HRSG	PIPE	70.482	22.479	593.62	-
12	Hot reheat steam at ST IP stop/control valve inlet	PIPE	ST	70.482	21.636	592.12	-
13	LP steam at HRSG outlet	HRSG	PIPE	7.255	3.229	225.83	-
14	LP steam at ST LP admission	PIPE	ST	7.255	3.011	224.83	-
15	IP turbine exhaust (crossover, internal)	ST	ST	70.482	2.974	320.25	-
16	LP turbine exhaust to condenser (UEEP)	ST	COND	77.737	0.02726	22.50	2,428.23
17	Condensate at hotwell outlet / CEP suction	COND	CEP	77.737	0.02726	22.45	-
18	Condensate at CEP discharge	CEP	CMIX	77.737	16.000	22.61	-
19	Fuel gas heater water return to condensate	FGH	CMIX	7.629	24.253	75.00	-
20	Condensate to HRSG condensate preheater inlet	CMIX	HRSG	85.367	15.500	27.32	-
21	Feedwater from LP drum to boiler feed pumps	HRSG	BFP	78.112	3.540	139.21	-
22	HP feedwater at BFP discharge to HP economiser	BFP	HRSG	58.028	118.581	141.12	-
23	IP feedwater at BFP interstage to IP economiser	BFP	HRSG	19.579	27.253	139.64	-
24	IP economiser water to performance fuel gas heater	HRSG	FGH	7.629	26.942	221.08	-
25	Circulating seawater to condenser inlet	SEA	COND	10,196.804	2.500	16.00	-
26	Circulating seawater from condenser outlet	COND	OUT	10,196.804	1.700	20.37	-
29	HP attemperation spray water (BFP discharge)	BFP	HRSG	0.505	118.581	141.12	-

3.5 Case SRC-LDO-100 - Site Reference Conditions, LDO with water injection, base load (guarantee PG-15/16)

Ambient 15.0 degC, 70.0 % RH, seawater 16.0 degC; GT 395.0 MW (100.0 % load), HR 8,650 kJ/kWh; ST 175.45 MW; GSU losses GT 982 kW, ST 472 kW; HRSG gas temperatures 622.0, 453.9, 360.3, 355.5, 292.2, 254.5, 200.9, 251.5, 164.9, 82.8 degC.

No	Description	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	AMB	GT	671.131	1.012	15.00	-
4	GT exhaust gas to HRSG inlet	GT	HRSG	715.820	1.041	622.00	-
5	Flue gas at HRSG outlet to stack	HRSG	STK	715.820	1.014	82.80	-
6	HP steam at HRSG outlet (HP SH outlet)	HRSG	PIPE	86.464	161.438	600.00	-
7	HP steam at ST main stop valve inlet	PIPE	ST	86.464	156.690	598.50	-
8	Cold reheat steam at HP turbine exhaust	ST	PIPE	86.464	35.290	371.24	-
9	Cold reheat steam at HRSG reheater inlet	PIPE	HRSG	86.464	34.697	370.24	-
10	IP superheated steam (joins cold reheat inside the HRSG)	HRSG	HRSG	17.220	35.595	318.67	-
11	Hot reheat steam at HRSG outlet	HRSG	PIPE	103.685	33.011	590.38	-
12	Hot reheat steam at ST IP stop/control valve inlet	PIPE	ST	103.685	31.769	588.88	-
13	LP steam at HRSG outlet	HRSG	PIPE	12.935	5.035	240.19	-
14	LP steam at ST LP admission	PIPE	ST	12.935	4.579	239.19	-
15	IP turbine exhaust (crossover, internal)	ST	ST	103.685	4.461	306.81	-
16	LP turbine exhaust to condenser (UEEP)	ST	COND	116.619	0.03246	25.40	2,357.33
17	Condensate at hotwell outlet / CEP suction	COND	CEP	116.619	0.03246	25.35	-
18	Condensate at CEP discharge	CEP	CMIX	116.619	16.000	25.51	-
20	Condensate to HRSG condensate preheater inlet	CMIX	HRSG	116.619	15.500	25.53	-
21	Feedwater from LP drum to boiler feed pumps	HRSG	BFP	103.685	5.669	156.58	-
22	HP feedwater at BFP discharge to HP economiser	BFP	HRSG	86.312	178.718	159.71	-
23	IP feedwater at BFP interstage to IP economiser	BFP	HRSG	17.220	38.671	157.21	-
25	Circulating seawater to condenser inlet	SEA	COND	10,196.804	2.500	16.00	-
26	Circulating seawater from condenser outlet	COND	OUT	10,196.804	1.700	22.33	-
27	LDO at liquid fuel skid inlet (IEC terminal point IF-02)	LDS	GT	22.123	5.000	30.00	-
28	Demineralised water for NOx water injection (IF-03)	DMW	GT	22.566	60.000	30.00	-
29	HP attemperation spray water (BFP discharge)	BFP	HRSG	0.152	178.718	159.71	-

3.6 Case WIN-NG-100 - Winter design (-4.8 degC, 99.6 %), natural gas, base load (GT at output limit)

Ambient -4.8 degC, 80.0 % RH, seawater 8.0 degC; GT 470.0 MW (100.0 % load), HR 8,131 kJ/kWh; ST 207.94 MW; GSU losses GT 1311 kW, ST 619 kW; HRSG gas temperatures 635.7, 447.0, 373.6, 368.9, 297.5, 263.3, 200.8, 260.5, 168.0, 77.0 degC.

No	Description	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	AMB	GT	810.099	1.012	-4.80	-
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	FGS	FGH	22.357	38.000	25.00	-
3	Heated fuel gas to GT combustion system	FGH	GT	22.357	35.500	215.00	-
4	GT exhaust gas to HRSG inlet	GT	HRSG	832.455	1.051	635.70	-
5	Flue gas at HRSG outlet to stack	HRSG	STK	832.455	1.014	76.96	-
6	HP steam at HRSG outlet (HP SH outlet)	HRSG	PIPE	103.345	192.084	591.64	-
7	HP steam at ST main stop valve inlet	PIPE	ST	103.345	186.382	590.14	-
8	Cold reheat steam at HP turbine exhaust	ST	PIPE	103.345	41.489	359.67	-
9	Cold reheat steam at HRSG reheater inlet	PIPE	HRSG	103.345	40.767	358.67	-
10	IP superheated steam (joins cold reheat inside the HRSG)	HRSG	HRSG	18.210	41.622	330.41	-
11	Hot reheat steam at HRSG outlet	HRSG	PIPE	121.556	38.796	594.97	-
12	Hot reheat steam at ST IP stop/control valve inlet	PIPE	ST	121.556	37.344	593.47	-
13	LP steam at HRSG outlet	HRSG	PIPE	12.389	5.599	251.78	-
14	LP steam at ST LP admission	PIPE	ST	12.389	5.232	250.78	-
15	IP turbine exhaust (crossover, internal)	ST	ST	121.556	5.124	305.90	-
16	LP turbine exhaust to condenser (UEEP)	ST	COND	133.945	0.02174	18.82	2,335.41
17	Condensate at hotwell outlet / CEP suction	COND	CEP	133.945	0.02174	18.77	-
18	Condensate at CEP discharge	CEP	CMIX	133.945	16.000	18.93	-
19	Fuel gas heater water return to condensate	FGH	CMIX	13.385	44.940	75.00	-
20	Condensate to HRSG condensate preheater inlet	CMIX	HRSG	147.330	15.500	24.09	-
21	Feedwater from LP drum to boiler feed pumps	HRSG	BFP	134.941	6.121	159.57	-
22	HP feedwater at BFP discharge to HP economiser	BFP	HRSG	103.345	215.283	163.41	-
23	IP feedwater at BFP interstage to IP economiser	BFP	HRSG	31.595	47.940	160.37	-
24	IP economiser water to performance fuel gas heater	HRSG	FGH	13.385	44.724	249.33	-
25	Circulating seawater to condenser inlet	SEA	COND	10,196.804	2.500	8.00	-
26	Circulating seawater from condenser outlet	COND	OUT	10,196.804	1.700	15.28	-

3.7 Case MIN-NG-100 - Minimum ambient for full capability (-8 degC), natural gas, base load

Ambient -8.0 degC, 80.0 % RH, seawater 6.9 degC; GT 470.0 MW (100.0 % load), HR 8,089 kJ/kWh; ST 204.92 MW; GSU losses GT 1311 kW, ST 604 kW; HRSG gas temperatures 634.0, 448.0, 372.5, 367.8, 297.0, 262.5, 200.2, 259.7, 167.4, 76.1 degC.

No	Description	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	AMB	GT	801.288	1.012	-8.00	-
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	FGS	FGH	22.242	38.000	25.00	-
3	Heated fuel gas to GT combustion system	FGH	GT	22.242	35.500	215.00	-
4	GT exhaust gas to HRSG inlet	GT	HRSG	823.529	1.050	634.04	-
5	Flue gas at HRSG outlet to stack	HRSG	STK	823.529	1.014	76.13	-
6	HP steam at HRSG outlet (HP SH outlet)	HRSG	PIPE	101.645	189.110	593.44	-
7	HP steam at ST main stop valve inlet	PIPE	ST	101.645	183.507	591.94	-
8	Cold reheat steam at HP turbine exhaust	ST	PIPE	101.645	40.872	361.39	-
9	Cold reheat steam at HRSG reheater inlet	PIPE	HRSG	101.645	40.163	360.39	-
10	IP superheated steam (joins cold reheat inside the HRSG)	HRSG	HRSG	18.132	41.023	329.42	-
11	Hot reheat steam at HRSG outlet	HRSG	PIPE	119.777	38.220	594.57	-
12	Hot reheat steam at ST IP stop/control valve inlet	PIPE	ST	119.777	36.789	593.07	-
13	LP steam at HRSG outlet	HRSG	PIPE	12.266	5.522	251.11	-
14	LP steam at ST LP admission	PIPE	ST	12.266	5.158	250.11	-
15	IP turbine exhaust (crossover, internal)	ST	ST	119.777	5.051	305.66	-
16	LP turbine exhaust to condenser (UEEP)	ST	COND	132.044	0.02013	17.59	2,335.98
17	Condensate at hotwell outlet / CEP suction	COND	CEP	132.044	0.02013	17.54	-
18	Condensate at CEP discharge	CEP	CMIX	132.044	16.000	17.70	-
19	Fuel gas heater water return to condensate	FGH	CMIX	13.398	44.331	75.00	-
20	Condensate to HRSG condensate preheater inlet	CMIX	HRSG	145.442	15.500	23.04	-
21	Feedwater from LP drum to boiler feed pumps	HRSG	BFP	133.176	6.041	159.05	-
22	HP feedwater at BFP discharge to HP economiser	BFP	HRSG	101.645	211.677	162.82	-
23	IP feedwater at BFP interstage to IP economiser	BFP	HRSG	31.530	47.331	159.84	-
24	IP economiser water to performance fuel gas heater	HRSG	FGH	13.398	44.135	248.35	-
25	Circulating seawater to condenser inlet	SEA	COND	10,196.804	2.500	6.90	-
26	Circulating seawater from condenser outlet	COND	OUT	10,196.804	1.700	14.10	-

3.8 Case SUM-NG-100 - Summer design (32.8 degC, 0.4 %), natural gas, base load

Ambient 32.8 degC, 45.0 % RH, seawater 24.0 degC; GT 372.8 MW (100.0 % load), HR 8,501 kJ/kWh; ST 184.53 MW; GSU losses GT 895 kW, ST 511 kW; HRSG gas temperatures 649.3, 455.9, 366.9, 362.3, 291.0, 256.8, 194.7, 254.1, 162.7, 81.7 degC.

No	Description	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	AMB	GT	710.980	1.012	32.80	-
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	FGS	FGH	18.541	38.000	25.00	-
3	Heated fuel gas to GT combustion system	FGH	GT	18.541	35.500	215.00	-
4	GT exhaust gas to HRSG inlet	GT	HRSG	729.521	1.042	649.26	-
5	Flue gas at HRSG outlet to stack	HRSG	STK	729.521	1.014	81.70	-
6	HP steam at HRSG outlet (HP SH outlet)	HRSG	PIPE	94.410	176.274	600.00	-
7	HP steam at ST main stop valve inlet	PIPE	ST	94.410	171.089	598.50	-
8	Cold reheat steam at HP turbine exhaust	ST	PIPE	94.410	37.942	367.31	-
9	Cold reheat steam at HRSG reheater inlet	PIPE	HRSG	94.410	37.283	366.31	-
10	IP superheated steam (joins cold reheat inside the HRSG)	HRSG	HRSG	15.807	37.988	328.31	-
11	Hot reheat steam at HRSG outlet	HRSG	PIPE	110.898	35.490	600.00	-
12	Hot reheat steam at ST IP stop/control valve inlet	PIPE	ST	110.898	34.168	598.50	-
13	LP steam at HRSG outlet	HRSG	PIPE	10.742	5.039	247.30	-
14	LP steam at ST LP admission	PIPE	ST	10.742	4.735	246.30	-
15	IP turbine exhaust (crossover, internal)	ST	ST	110.898	4.653	309.17	-
16	LP turbine exhaust to condenser (UEEP)	ST	COND	121.641	0.05271	33.82	2,396.36
17	Condensate at hotwell outlet / CEP suction	COND	CEP	121.641	0.05271	33.77	-
18	Condensate at CEP discharge	CEP	CMIX	121.641	16.000	33.95	-
19	Fuel gas heater water return to condensate	FGH	CMIX	11.418	39.812	75.00	-
20	Condensate to HRSG condensate preheater inlet	CMIX	HRSG	133.059	15.500	37.53	-
21	Feedwater from LP drum to boiler feed pumps	HRSG	BFP	122.317	5.476	155.24	-
22	HP feedwater at BFP discharge to HP economiser	BFP	HRSG	92.894	195.618	158.66	-
23	IP feedwater at BFP interstage to IP economiser	BFP	HRSG	27.226	42.812	155.95	-
24	IP economiser water to performance fuel gas heater	HRSG	FGH	11.418	40.810	244.84	-
25	Circulating seawater to condenser inlet	SEA	COND	10,196.804	2.500	24.00	-
26	Circulating seawater from condenser outlet	COND	OUT	10,196.804	1.700	30.61	-
29	HP attemperation spray water (BFP discharge)	BFP	HRSG	1.516	195.618	158.66	-
30	Reheat attemperation spray water (BFP interstage)	BFP	HRSG	0.681	42.812	155.95	-

3.9 Case MAX-NG-100 - Maximum ambient (40 degC) with maximum seawater (27 degC), natural gas, base load

Ambient 40.0 degC, 35.0 % RH, seawater 27.0 degC; GT 348.2 MW (100.0 % load), HR 8,595 kJ/kWh; ST 175.84 MW; GSU losses GT 805 kW, ST 474 kW; HRSG gas temperatures 653.0, 458.3, 364.0, 359.4, 288.7, 254.4, 192.5, 251.7, 160.7, 81.9 degC.

No	Description	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	AMB	GT	678.416	1.012	40.00	-
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	FGS	FGH	17.510	38.000	25.00	-
3	Heated fuel gas to GT combustion system	FGH	GT	17.510	35.500	215.00	-
4	GT exhaust gas to HRSG inlet	GT	HRSG	695.926	1.039	653.00	-
5	Flue gas at HRSG outlet to stack	HRSG	STK	695.926	1.014	81.92	-
6	HP steam at HRSG outlet (HP SH outlet)	HRSG	PIPE	91.085	170.065	600.00	-
7	HP steam at ST main stop valve inlet	PIPE	ST	91.085	165.063	598.50	-
8	Cold reheat steam at HP turbine exhaust	ST	PIPE	91.085	36.635	368.13	-
9	Cold reheat steam at HRSG reheater inlet	PIPE	HRSG	91.085	36.000	367.13	-
10	IP superheated steam (joins cold reheat inside the HRSG)	HRSG	HRSG	15.050	36.661	327.03	-
11	Hot reheat steam at HRSG outlet	HRSG	PIPE	107.082	34.268	600.00	-
12	Hot reheat steam at ST IP stop/control valve inlet	PIPE	ST	107.082	32.992	598.50	-
13	LP steam at HRSG outlet	HRSG	PIPE	10.156	4.840	245.65	-
14	LP steam at ST LP admission	PIPE	ST	10.156	4.558	244.65	-
15	IP turbine exhaust (crossover, internal)	ST	ST	107.082	4.485	309.46	-
16	LP turbine exhaust to condenser (UEEP)	ST	COND	117.238	0.06113	36.50	2,416.50
17	Condensate at hotwell outlet / CEP suction	COND	CEP	117.238	0.06113	36.45	-
18	Condensate at CEP discharge	CEP	CMIX	117.238	16.000	36.63	-
19	Fuel gas heater water return to condensate	FGH	CMIX	10.917	38.089	75.00	-
20	Condensate to HRSG condensate preheater inlet	CMIX	HRSG	128.155	15.500	39.95	-
21	Feedwater from LP drum to boiler feed pumps	HRSG	BFP	118.000	5.246	153.61	-
22	HP feedwater at BFP discharge to HP economiser	BFP	HRSG	89.026	188.077	156.87	-
23	IP feedwater at BFP interstage to IP economiser	BFP	HRSG	25.967	41.089	154.28	-
24	IP economiser water to performance fuel gas heater	HRSG	FGH	10.917	39.404	242.91	-
25	Circulating seawater to condenser inlet	SEA	COND	10,196.804	2.500	27.00	-
26	Circulating seawater from condenser outlet	COND	OUT	10,196.804	1.700	33.40	-
29	HP attemperation spray water (BFP discharge)	BFP	HRSG	2.058	188.077	156.87	-
30	Reheat attemperation spray water (BFP interstage)	BFP	HRSG	0.948	41.089	154.28	-